Assessment Part 1: QnA

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Long Answer No 2:

There are some common challenges to deploying machine learning systems into production, especially in areas such as e-commerce that change rapidly. Data drift, latency/scalability, and monitoring gaps are three important issues.

Input distributions change over the time and/or feature-target relationships change over time (data drift or concept drift, e.g., seasonal behavior changes or user preferences). Model accuracy quietly declines if not solved, resulting in lower conversions, suboptimal recommendations and suboptimal operations. Practical measures such as statistical monitoring, safe deployment using shadow deployments, and automatic retraining based on performance limits or drift detection. There are trade-offs between responsiveness and stability, such as retraining more often to improve accuracy, but that can raise compute expenses and risk of overfitting, and can also delay adaptation to real-world changes; conservative thresholds can improve reliability, but can cause slower adaptation to a change in reality.

During traffic peaks, latency and scalability become important to ensure that inference meets the strictest Service Level Objective (SLOs) (e.g., under 100ms) and that the user experience is not compromised. If it is not handled properly, the delays on the pages, checkout failures and high infrastructure cost. These solutions include model optimization, multi-tier caching, async inference for non-critical predictions, and auto-scaling infrastructure and fallback to heuristic rules during outages. There are compromises between speed and accuracy: optimization techniques can slightly reduce the quality of the models, aggressive caching can lead to stale predictions, and fallback can compromise the personalization of the prediction.

Monitoring gaps are due to the fact that there are failures in the traditional infrastructure that are not captured by traditional metrics, such as silent accuracy loss, biased prediction or feedback loops. Systems can make recommendations for unavailable items or highlight bias if they are not observed properly. Useful techniques are those that deal with data quality, such as validation; online business metrics (CTR, revenue); sampling explainability tools for important predictions; and end-to-end tracing for debugging. The downside: The granularity of monitoring provides greater detection, but contributes to storage and engineering overhead.

Cross functional cooperation enhances reliability. Data scientists set the modelling goals, engineers build a solid serving and monitoring solution, and product managers ensure that everything is aligned with the impact on the user and business objectives. The dashboards are shared, there are clear escalation protocols and regular syncs prevent silos, and allows for coordinated response to issues such as drift alerts or performance regressions.

These are complemented by modern tools. Automation frameworks standardize retraining pipelines, feature stores provide consistency for training/serving, and ML-specific monitoring platforms offer observability. LLM systems can be used for anomaly detection, alert summarization. The integration complexity and maintenance overhead are added by tooling, however. The best approach is to begin with critical paths, instrument incrementally and scale up practices as the system grows. In the end, effective ML systems rely on a mix of technical solutions, collaborative approaches, and pragmatic implementation of tools, fostering iterative feedback that aligns with business requirements.

Short Answer No 1.1

Overfitting is a situation where a model becomes too specialized in order to fit the training data, picking up the noise and not the underlying patterns, so it performs well on the training set, but not so well on the test set. Underfitting occurs when the model is oversimplified and cannot capture the underlying relationships, leading to low accuracy scores on training and test sets. Both have a negative effect on generalization: overfitted models are not robust to new inputs and underfitted models have high bias, and cannot represent the real world complexity. A high-degree polynomial regressor that is able to remember all the training points will make wild predictions for new points, while a straight line fitted to data that is very non-linear will consistently miss important trends. The key to ideal generalization is a balance of model capacity and the complexity of the data, with the learned function reflecting the signal in the data and not memorizing noise or missing structure.

Short Answer No 2.1

CNNs are used to process grid-structured data, such as images, and RNNs are used to process sequential data, such as text. From an architectural perspective, CNNs learn spatial features from using sliding filters that have shared weights. In RNNs, the data is passed sequentially through the network, the hidden state is carried forward and shared across time steps, allowing the network to learn temporal dependencies. CNNs propagate information locally to globally, while RNNs propagate information sequentially. CNNs are used in image classification and medical imaging. RNNs are used in language modelling and speech recognition.

Vanishing/exploding gradients are a problem in training deep networks, which means the signals get weaker or stronger as they pass through the different layers of the network. Gradient clipping and gated architectures (LSTMs) overcome this. Overfitting is a situation in which models remember what they have been fed instead of learning. Some regularizing methods for mitigating include dropout or early stopping or batch normalization to stop training when the best validation performance is achieved. These strategies ensure stable and generalizing deep learning systems together.